Unit – V: Working with React Framework

5.1 Event Handling in React

What is Event Handling?

Event handling means **performing an action when a user interacts with the webpage**.

Examples of events:

- Clicking a button
- Typing in an input box
- Submitting a form
- Moving the mouse

React handles events similar to JavaScript but uses **camelCase syntax**.

Example:

```
<button onClick={handleClick}>Click Me</button>
```

Example Program

```
function App() {

  function showMessage() {
    alert("Button Clicked!");
  }

  return (
    <div>
      <button onClick={showMessage}>Click Here</button>
    </div>
  );
}
```

Output:
When the button is clicked → **Message appears**.

Binding Event Handlers

Binding means **connecting the event with a function**.

Example:

```
<button onClick={handleClick}>Submit</button>
```

Here:

- onClick → Event
- handleClick → Function

Arrow Functions vs Regular Functions

In React, we can create functions using regular functions or arrow functions.

Regular Function

```
function show() {
  console.log("Hello");
}
```

Arrow Function

```
const show = () => {
  console.log("Hello");
};
```

Difference

Regular Function	**Arrow Function**
Uses function keyword	Uses => symbol
Needs binding in class components	No binding required
More lines of code	Short and simple

Example in React:

```
<button onClick={() => alert("Hello")}>Click</button>
```

5.2 Working with Forms

Forms are used to **collect user input** in a web application.

Examples:

- Login form
- Registration form
- Contact form

Adding Components in Forms

React form elements include:

- Input
- Textarea
- Select
- Button

Example:

```
<input type="text" placeholder="Enter Name" />
```

Handling Form

React uses **state to manage form data**.

Example:

```
const [name, setName] = useState("");

<input
 type="text"
 value={name}
 onChange={(e) => setName(e.target.value)}
/>
```

Here:

- name → state variable
- setName → update function

Submitting Forms

Form submission is handled using **onSubmit event**.

Example:

```
function handleSubmit(e) {
  e.preventDefault();
  alert("Form Submitted");
}

<form onSubmit={handleSubmit}>
```

preventDefault() prevents page refresh.

Form Validation

Validation checks whether user input is **correct or not**.

Examples:

- Required fields
- Valid email format
- Password length

Example:

```
if(name === ""){
  alert("Name is required");
}
```

5.3 Lists and Keys

Lists are used to **display multiple items dynamically**.

Example:

- Student list
- Product list
- Menu items

Rendering Lists

React can display lists using **arrays**.

Example:

```
const fruits = ["Apple", "Mango", "Banana"];

function App() {
  return (
    <ul>
      { fruits.map((fruit) => (
        <li>{fruit}</li>
      ))}
    </ul>
  );
}
```

Output:

Apple
Mango
Banana

List with Key

Keys help React **identify each item in the list**.

Example:

```
const fruits = ["Apple", "Mango", "Banana"];

<ul>
  {fruits.map((fruit, index) => (
    <li key={index}>{fruit}</li>
  ))}
</ul>
```

key should be **unique**.

Using map() to Render Lists

map() function is used to **loop through array elements**.

Example:

```
const numbers = [1,2,3,4];

numbers.map((num) => {
  return <li>{num}</li>;
});
```

Output:

1
2
3
4

5.4 Cascading Style Sheets (CSS)

CSS is used to **style and design web pages**.

It controls:

- Colors
- Fonts
- Layout
- Spacing

Types of Style Sheets

1 Inline CSS

Style written inside the element.

Example:

```
<h2 style={{color:"red"}}>Hello</h2>
```

---

2 Internal CSS

Style written inside <style> tag.

Example:

```
<style>
h1 {
  color: blue;
}
</style>
```

---

3 External CSS

Style written in a separate file.

Example:

```
/* style.css */
h1{
 color: green;
}
```

Import in React:

```
import "./style.css";
```

Styling Libraries

Styling libraries provide **ready-made components and styles**.

Examples:

- Bootstrap
- Material UI
- Tailwind CSS

Advantages:

- Faster development
- Responsive design
- Pre-built UI components

Popular CSS Frameworks

Bootstrap

Bootstrap is a popular CSS framework used to create responsive websites.

Example:

```
<button class="btn btn-primary">
Click
</button>
```

Bootstrap provides:

- Buttons
- Forms
- Grid system
- Navigation bars

Material UI

Material UI is a **React component library** based on Google's Material Design.

Example:

```
import Button from '@mui/material/Button';

<Button variant="contained">Click</Button>
```

Unit V Summary

Topic	Description
Event Handling	Handling user actions like click
Arrow Functions	Short syntax for functions
React Forms	Used to collect user input
Form Validation	Checking input correctness
Lists & Keys	Display multiple items
map()	Loop through array
CSS	Styling web pages
Bootstrap	Popular CSS framework
Material UI	React UI library